

PowerPoint Presentation: Chapter 1 - Pneumonia Epidemiology & Pathophysiology

Dr. Siddalingaiah H S - MBBS Phase 3 Part 1 Integration Class

Slide 1: Title Slide

Pneumonia Treatment & Immunization CBME Module Chapter 1: Introduction to Pneumonia - Epidemiology & Pathophysiology

Dr. Siddalingaiah H S Professor, Community Medicine MBBS Phase 3 Part 1 Integration Class - Day 1

Slide 2: Learning Objectives

By end of this session, MBBS students will be able to:

1. Define pneumonia using clinical & radiological criteria
2. Classify pneumonia types by etiology & acquisition
3. Explain global & national pneumonia burden data
4. Describe pathophysiological mechanisms of pneumonia
5. Identify host & environmental risk factors

CBME Competencies: Medical Knowledge, Patient Care, Systems-Based Practice

Slide 3: What is Pneumonia?

Acute infection of lung parenchyma characterized by:

- **Consolidation of alveoli**
- **Inflammatory exudate accumulation**
- **Impaired gas exchange**
- **Typical vs Atypical presentations**
- **Community vs Hospital-acquired**

Visual: Chest X-ray showing lobar consolidation

Slide 4: Pneumonia Classification

Category	Definition	Examples	Microbiology
Community-Acquired	<48 hours hospital admission	Healthy adults	<i>S. pneumoniae</i> , <i>M. pneumoniae</i>
Hospital-Acquired	≥48 hours admission	ICU patients	MDR pathogens, <i>P. aeruginosa</i>

Category	Definition	Examples	Microbiology
Ventilator-Associated	≥48-72 hours intubation	Mechanically ventilated	ESBL producers, MRSA
Healthcare-Associated	Recent healthcare exposure	LTCF residents	Mixed bacterial flora

Slide 5: Global Pneumonia Burden

 Key Statistics:

- **Annual cases:** 156 million worldwide
- **Annual deaths:** 2.5 million people
- **Leading cause:** Top infectious disease killer
- **Age distribution:** 71% in children <5 years
- **Adult mortality:** 600,000 deaths annually

World Map highlighting pneumonia burden by region

Slide 6: Indian Pneumonia Statistics

IN Indian Context:

- **Annual episodes:** 40-50 million cases
- **Mortality rate:** 340,000 deaths annually
- **Case fatality:** 2.3 per 1,000 persons
- **Child mortality:** 17% of under-5 deaths
- **Highest burden:** Uttar Pradesh, Bihar, Rajasthan

Indian map showing state-wise pneumonia burden

Slide 7: Etiological Agents

Microbial Spectrum in CAP:

1. **Streptococcus pneumoniae** - 25-40% (most common bacterial)
2. **Mycoplasma pneumoniae** - 15-30% (school-age peak)
3. **Chlamydophila pneumoniae** - 10-15% (young adults)
4. **Haemophilus influenzae** - 5-15% (COPD patients)
5. **Legionella pneumophila** - 2-10% (water systems)
6. **Respiratory viruses** - 15-20% (winter epidemics)
7. **Staphylococcus aureus** - 2-5% (post-viral)

Pie chart showing distribution of pathogens

Slide 8: Risk Factors for Pneumonia

Host Factors:

- **Age extremes:** <2 years and >65 years
- **Underlying diseases:** COPD, diabetes, heart failure
- **Immunodeficiency:** HIV, malignancy, immunosuppression
- **Lifestyle:** Smoking, alcoholism, poor nutrition

Environmental Factors:

- **Air pollution:** Indoor smoke, particulate matter
 - **Crowding:** Urban slums, dormitories
 - **Malnutrition:** Vitamin A, zinc deficiency
 - **Seasonal:** Winter epidemics, monsoon peaks
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Slide 9: Pathophysiology of Pneumonia

Four-Step Sequence:

1. **Colonization** - Pathogen adheres to epithelium
2. **Invasion** - Bacterial multiplication in alveoli
3. **Inflammation** - Cytokine release, neutrophil migration
4. **Consolidation** - Exudate fills air spaces, impairs gas exchange

Animated diagram showing pathogenesis sequence

Slide 10: Host Defense Mechanisms

Anatomical Barriers:

- **Upper airways:** Nasal filtration, mucociliary clearance
- **Cough reflex:** Prevents aspiration
- **Bronchial tree:** Hair cells, mucus production

Cellular Immunity:

- **Alveolar macrophages:** Phagocytosis
- **Neutrophils:** Acute inflammatory response
- **Complement system:** Opsonization, chemotaxis

Humoral Immunity:

- **Antibodies:** Serotype-specific protection
 - **Immunoglobulins:** IgG, IgM, IgA
 - **Memory cells:** Long-term immunity
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Slide 11: Aspiration vs Inhalation Routes

Aspiration Pneumonia:

- **Most common route:** Oropharyngeal contents
- **Risk factors:** CNS disorders, dysphagia, unconsciousness
- **Anaerobic bacteria:** Predominant in elderly
- **Chemical pneumonitis:** Gastric acid injury

Inhalation Pneumonia:

- **Droplet nuclei:** Mycobacteria, viruses, Legionella
- **Airborne transmission:** RSV, influenza aerosols
- **Smaller inoculum:** Survives upper airway defenses
- **Diffuse involvement:** Often bilateral distribution

Slide 12: Typical vs Atypical Pneumonia

Feature	Typical Pneumonia	Atypical Pneumonia
Onset	Acute, sudden	Gradual, subacute
Fever	High, shaking chills	Low-grade
Cough	Productive, purulent	Dry, persistent
Chest pain	Pleuritic	Minimal
X-ray	Lobar consolidation	Interstitial pattern
White cells	Elevated neutrophils	Normal/mild elevation

Side-by-side X-ray comparisons

Slide 13: Clinical Manifestations

Respiratory Symptoms:

- **Dyspnea:** Progressive shortness of breath
- **Cough:** Dry → productive with purulent sputum
- **Chest pain:** Pleuritic, worse on inspiration
- **Hemoptysis:** Rusty sputum (*S. pneumoniae*)

Systemic Symptoms:

- **Fever:** High-grade with chills/rigors
 - **Fatigue:** Generalized weakness
 - **Anorexia:** Loss of appetite
 - **Myalgias:** Muscle aches and pains
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Slide 14: Complications of Pneumonia

Respiratory Complications:

- **Pleural effusion/empyema**
- **Lung abscess**
- **Pneumothorax**
- **ARDS**

Systemic Complications:

- **Sepsis/septic shock**
- **Acute kidney injury**
- **Heart complications**
- **Metabolic disorders**

Slide 15: CBME Assessment Questions

Question 1: Most common cause of CAP? A) Haemophilus influenzae B) Streptococcus pneumoniae ★ C) Mycoplasma pneumoniae D) Legionella pneumophila

Question 2: Pneumonia mortality in India? A) 10,000 deaths/year B) 50,000 deaths/year C) 340,000 deaths/year ★ D) 1 million deaths/year

Slide 16: Learning Activities

In-Class Activities:

1. **Case Analysis:** Review 3 pneumonia patient scenarios
2. **Risk Factor Mapping:** Group exercise on local determinants
3. **Microbiology Quiz:** Identify pathogens from descriptions
4. **Epidemiology Exercise:** Calculate local pneumonia burden

Homework Assignment:

- Recall one pneumonia case from recent clinical posting
- Identify risk factors present in that patient
- Research local pneumonia statistics for tomorrow's discussion

Slide 17: Key Takeaways

Remember These Points:

1. **Pneumonia = Acute lung infection with consolidation**
2. **2.5 million deaths globally - mostly preventable**
3. **S. pneumoniae most common bacterial cause**
4. **Host + environmental factors determine risk**
5. **Typical vs atypical clinical presentations differ**
6. **Vaccination prevents most vaccine-preventable pneumonia**

Slide 18: References & Further Reading

Key References:

1. WHO Pneumonia Fact Sheet (2023)
2. Indian Academy of Pediatrics Guidelines (2022)
3. ATS/IDSA CAP Guidelines (2019)
4. Mandell LA, et al. Clinical Infectious Diseases (2019)

Useful Links:

- WHO Pneumonia: <https://www.who.int/health-topics/pneumonia>
- IAP Guidelines: <https://iapindia.org/guidelines/>
- CDC Pneumonia: <https://www.cdc.gov/pneumonia/>

Slide 19: Session Evaluation

Please provide feedback:

What was most useful? _____

What needs improvement? _____

Additional topics to cover? _____

Overall rating: ★ ★ ★ ★ ★

Slide 20: Q&A Session

Questions & Discussion

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Next Session: Chapter 2 - Pneumonia Classification (CAP vs HAP) **Date:** [Insert date] | **Time:** 9:00 AM - 10:30 AM **Venue:** Integration Class Hall

Total Slides: 20 Estimated Duration: 45-60 minutes Faculty: Dr. H S Lingaiah Target Audience: MBBS Phase 3 Part 1 Students